Coherence Under Decay: A Shared Control Template for Qubits and a Zero-Debt Commons

Merging the Ariel Quantum Boundary Proposal with The Zero-Point Chronomimetic Framework (v2.0)

Prepared for: Ryan Espinosa · **Date:** November 12, 2026 · **AI assistant:** Grok Bot

CONCEPTUAL FRAMEWORK. This is not validated quantum theory and not validated economics. It is an analogy-driven mapping from a personal control framework onto laboratory observables and app-level behavioral metrics. It is offered as a **research agenda, not a claim of new physics or a proven economic model.** The physics parts are standard textbook results (labelled standard result). The economic and behavioral parts are design hypotheses to be tested. **The link between the two is a structural analogy, a shared control-theory template. It does not claim that economics is quantum mechanics.**

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0. Reader's guide and the unifying template

This document joins two of Ryan's works: the revised quantum boundary proposal (Part I) and The Zero-Point Chronomimetic Framework v2.0 (Part II). Part III shows exactly where they share mathematical **form**, and where that shared form stops.

The unifying idea is named here the **Coherence-Under-Decay (CUD) template**:

> Keep a system useful and coherent while it is pulled toward decay by idleness, noise, and uncontrolled coupling. Use bounded, gentle control, and always keep a protected rest state that is reachable, honest, and never "empty."

Both a qubit and a healthy commons can be described as instances of this template. That is a statement about **shared control structure**, the same way a pendulum and an RLC circuit obey the same differential equation without being the same thing.

Three layers of meaning. Each claim below is tagged as one of:

- **[Empirical]**: measurable, testable, and could turn out false.
- **[Design]**: an engineering or product choice, judged by outcomes.

- **[Values]**: Ryan's faith and meaning framework (Circuit of Grace, living with Source, Kairos). It sets **what is worth optimizing**. It is not tested by experiment and is not diminished by being labelled.

PART I: The Quantum Layer (condensed)

1. Three separate physical layers

The earlier proposal fixed four overclaims in the original draft:

- Cooling does **not** remove zero-point energy.
- Dividing by (T→0) is singular.
- QED UV regularization is a calculational step, not stopping power.
- A metaphorical "gate" is not a gate voltage.

The physics is kept as three layers that **add as error rates**:

Layer	Mechanism	Measurable control	Ariel label
A. Coherent boundary control	Tunneling / avoided crossings	θ ∈ {Φ_ext, n_g, drive phase}	B_gate
B. Thermal-vacuum referencing	Mode occupation n̄ → 0, **½ħω remains**	Temperature T, cooling, reset	"freeze-to-baseline"
C. Bath decoupling	Filtering, Purcell filters, shielding, dynamical decoupling	Shielding/filter stack □	S_stop, E_core (as Δ_SC)

```
EQUATIONS (all standard results)
  A  H = 4E_C(n − n_g)² − E_J(Φ_ext) cos φ ;   E_J(Φ) = E_J,max |cos(πΦ/Φ_0)|
     P_LZ = exp(−πΔ²/(2ħα))                    (Landau–Zener, H = ½[αt σ_z + Δ σ_x])
  B  x = ħω/(k_B T) ;  n̄ = 1/(e^x − 1) → 0 ;  ⟨E⟩ = ħω(n̄ + ½) → ½ħω  (zero-point NOT
removed)
     1/T_1 = Γ_1⁰ (2n̄ + 1)
  C  J_eff(ω) = |H_filter(ω)|² J(ω) ;  Γ_P ≈ κ (g/Δ_qr)² ;  Γ_qp ∝ x_qp
  Fidelity (amplitude damping + pure dephasing, t_g ≪ T_1,T_2):
     F̄ ≈ 1 − (t_g/3)(Γ_1 + Γ_φ)

PARAMS
  Λ_quantum ↦ E_zp = ½ħω_q (fixed, not cosmological Λ)
  S_stop    ↦ shielding/filter stack □; dE/dx used only for the ionizing-radiation
budget
  E_core    ↦ superconducting gap Δ_SC (speculative mapping)
  I_smart   ↦ the optimizer itself
```

Quantum objective [Empirical/Design]:

```
max over {θ(·), T, □}   F̄ ≈ 1 − (t_g/3)[Γ_1 + Γ_φ] − ε_ctrl[θ(·)]
  Γ_1 = Γ_1⁰(□)(2n̄(ω_q,T) + 1) + Γ_P(□) + Γ_qp(□)
  s.t. n̄ ≤ n_max ;  Γ_1 + Γ_φ ≤ Γ_max ;  U[θ] = U* ± ε_target ;  |θ| ≤ θ̇_max
```

Key references: Caldeira & Leggett (1981); Devoret, Martinis & Clarke (1985); Zener (1932); Koch et al. (2007); Diedrich et al. (1989); Safavi-Naeini et al. (2012); Blais et al. (2004, 2021); Houck et al. (2008); Viola, Knill & Lloyd (1999); Vepsäläinen et al. (2020); McEwen et al. (2022); Peskin & Schroeder (1995) for the contrast between regularization and experiment.

One physics lesson carries into Part II: no system reaches perfect isolation or perfect zero. A real qubit always has residual Γ, and its ground state keeps ½ħω. Part II applies this lesson to the Sanctuary and to "friction → 0".

PART II: The Zero-Point Economic and Behavioral Layer (revised)

2. Postulates: reinforced, challenged, and re-tagged

P1. Circuit of Grace (non-extractive coherence). [Values] Value as reciprocal giving rather than hoarding, and living with Source in place of scarcity anxiety. These are kept intact as the framework's purpose. [Design] The zero-debt ledger (balances ≥ 0, no loans, no interest) is a clear, implementable rule. Challenge: the symbol **f** is used both for "debt" (f = 0) and for "friction". Use **d** for debt and **f** for friction. Tension: the axiom "information is free" sits uneasily with charging Chrons for modules and locking resale. A consistent option is free reading access, with Chrons used for acknowledgment, curation, and derivative publishing rights.

P2. Polarity of Chaos and Rest (Ingress–Egress). [Values/Design] Rest gains meaning by contrast with activity. This is a reasonable design principle, and it echoes Sabbath rhythms within Ryan's own tradition. [Empirical, testable] Does a structured activity→rest cycle improve wellbeing or reduce compulsive use compared with no cycle? That is a pilot question (Section 9), not a law.

P3. Asymptotic equality of time and action. This is a **normative aspiration, not a guaranteed outcome.** Equality holds only if the minting rule is time-indexed and capped, so that early or prolific users cannot compound an advantage. The closest real precedent is **time banking** (Edgar Cahn's Time Dollars), where one hour of service earns one credit regardless of the service's market price. Time banks show the idea can work at community scale. They also show the usual challenges: keeping participants engaged and measuring contribution fairly. Cross-check: the passive yield on held vault modules (Section 4.1 of v2.0) rewards holding. It works against both demurrage and P3. I recommend removing passive yield or making it strictly smaller than demurrage.

P4. Demurrage of isolation. [Design, with historical precedent] V(t) = V₀e^{−λt} is sound. Silvio Gesell proposed Freigeld (money that loses value over time, implemented as "stamp scrip") in Die natürliche Wirtschaftsordnung (1916). In the town of Wörgl, Austria, in 1932–33, Mayor Michael Unterguggenberger issued labor certificates that required a monthly stamp of about 1% of face value to stay valid. Local circulation and public works increased, and the Austrian National Bank halted the scheme in 1933. Irving Fisher (Stamp Scrip, 1933) and Keynes (General Theory, 1936, ch. 23) discussed Gesell seriously. Treat Wörgl as a short, suggestive episode, not a controlled experiment. Cross-check: in v2.0, the Sanctuary freezes demurrage (λ → 0) with no limit. That makes it a perfect hoarding shelter and defeats P4. Fix: give the Sanctuary a small residual rate λ_s > 0, or cap staking duration. This is the economic counterpart of "no perfect isolation" in Part I.

P5. Patience over impulsion (Honor Engine). [Design] Rewarding contemplation instead of speculation is a worthy inversion. Challenges:

- A **lottery** is a variable-ratio reward, which is the reinforcement schedule most associated with persistent, compulsive responding (Ferster & Skinner 1957; Schüll 2012).
- Many jurisdictions define an illegal lottery as prize + chance + consideration. Get legal review. Keeping entry free and avoiding cash-convertible prizes reduces the risk.
- If patience weight accrues for time spent in the app's Sanctuary, it rewards screen time, which contradicts "clean egress."
- Fixes: count time away from the app as stillness, and prefer deterministic recognition such as rotating stewardship or equal shares to the treasury. If a draw is kept, keep it infrequent, without animations, without near-miss displays, and published transparently. The sortition literature (Goodwin 1992; Stone 2011) gives defensible fairness rationales for lotteries used to allocate roles. It does not cover lotteries used as engagement rewards.

3. Fixed equations

All economic quantities below are **[Design]** definitions that need calibration in a pilot. Thresholds are not known in advance and must not be invented.

3.1 Circuit of Grace circulation (replaces T_grace)

The original T_grace = ∮ Contribution/(Friction→0) dt **diverges**: any nonzero contribution divided by friction approaching zero is infinite. The ∮ also implies a closed loop that is not defined.

```
EQUATIONS
  Circulation velocity (finite, measurable):
     G_W = F_W / (W · B̄_W)                          [1/day]
  Friction-adjusted grace flow (optional):
     G_f = F_W / (W · B̄_W) · 1/(1 + f/f_min)         (finite; equals G_W when f = 0)

PARAMS
  F_W  = total Chrons transferred to *other* participants or the treasury in window W
         (self-transfers excluded)
  B̄_W  = mean total balance held over W ;  W = window length (days)
  f    = measured friction (fee fraction or median seconds of overhead per transfer)
  f_min > 0 = friction floor (a real system is never exactly frictionless)
```

G_W is the velocity of money (Fisher's MV = PT) applied to Chrons. **Higher is not always better**: very high velocity can signal churn or compulsion. So the objective in Section 7 uses a target band, not a maximum.

3.2 Chaos index (dimensionless, bounded)

```
  C_chaos = ṽ · H̃ · 1/(1 + T_stab/τ_0)            ∈ [0, 1]
  ṽ  = min(1, |Δv|/v_ref)         normalized change in transaction velocity
  H̃  = H/log₂N,  H = −Σ p_i log₂ p_i   Shannon entropy of transactions across N
categories
                                    or counterparties (Shannon 1948)
  T_stab = recent stabilization (rest) time ;  τ_0 = reference time scale
```

This makes "entropy" **measurable**. Anywhere else in the framework, "entropy" and "thermodynamic" are **metaphors** unless defined this way.

3.3 Entropic decay and diminishing idle returns

```
  V(t) = V₀ e^{−λ t_idle} ;  half-life t½ = ln2/λ          (exponential demurrage)
  D(t) = 1/(1 + λ_h t_idle)                                 (hyperbolic efficiency
curve)
```

About D(t): this is the same functional form as **hyperbolic discounting**, V = A/(1 + kD) (Ainslie 1975; Mazur 1987; quasi-hyperbolic form in Laibson 1997). Two cautions apply.

- In the app, D(t) is a yield-efficiency schedule set by the designer as a function of idle time. It is not a model of how users discount future rewards. Only the mathematical form is shared.
- Exponential decay has a constant hazard rate. Hyperbolic decay falls fast at first and then flattens. Pairing them is fine, but λ and λ_h should be different, named parameters.

3.4 Danger threshold monitor (no division by zero)

The original D_danger = (velocity × impulsive frequency)/(Sanctuary duration) is infinite for any user who has never entered the Sanctuary, including every new user.

```
  Ratio form:     D_danger = ṽ_tx · r̃_imp / (1 + s/s_0)            ≥ 0, finite
  Logistic form:  D_danger = σ( a·ṽ_tx + b·r̃_imp − c·s/s_0 − d ),   σ(z) = 1/
(1+e^{−z}) ∈ (0,1)

PARAMS
  ṽ_tx  = transactions per hour / reference rate
  r̃_imp = impulsive purchases per hour / reference rate
          (impulsive := purchase < k seconds after first view, or ≥ m purchases in M
minutes;
           k, m, M set in the pilot)
  s     = rest time over the window, including time *away from the app* ;  s_0 = scale
  a, b, c, d, D_crit = calibrated from pilot data, not chosen a priori
```

3.5 Semantic delta (anti-arbitrage)

ΔH = 1 − cos(V_orig, V_mod), computed on sentence embeddings (e.g. Reimers & Gurevych 2019), is a legitimate similarity measure. **It is also gameable**: padding, paraphrase, or appending unrelated text can raise ΔH without adding insight.

Recommendations:

- Use ΔH only as a gate to human or peer review, not as automatic approval.
- Compare against **all** of the module's sources.
- Cap the credit given to appended text.

Loophole in v2.0: Composite Bundles unlock automatically. Bundling two unmodified modules bypasses the resale lock. Fix: apply ΔH to the bundle against the maximum similarity to any of its components.

4. App specification: summary and safety alignment

Modules kept: Information Commons marketplace, Research Vault, Composite Bundle Creator, Decay Simulator, Action Generator, and Contribution Portal, as specified in v2.0. The zero-debt ledger is kept.

Safety features kept and strengthened. These are tied to MindFrame/Ariel principles: no reassurance loops, private by default, and no gamified compulsion.

v2.0 feature	Keep	Strengthen with
Danger monitor (emerald → indigo)	Yes	Offer reflection; never punish. No escalating alarms that become a reassurance loop. Thresholds come from the pilot.
Sanctuary	Yes	Residual λ_s > 0; patience credited for time off-app; no streaks

v2.0 feature	Keep	Strengthen with
Clean egress	Yes, central	Single-tap exit, state saved, **no** "you'll lose X" messages, no re-engagement push notifications by default
2-second decay ticker	Revise	Compute on open instead of animating continuously. Live ticking counters invite compulsive checking.
Patience lottery	Revise	Prefer deterministic recognition; if a draw is kept, see P5 constraints
Stochastic concept seeding	Keep, carefully	No slot-machine reveal animation
Telemetry	Revise	Private by default, stored on device where possible, explicit opt-in for any research data

Data integrity. In the sample JSON, the author **"Dr. Alistair Vance"** appears to be a placeholder. Fictional authors should not be shown as real, so use "Example Author" or the real contributor's name. Likewise, ratings such as `"verification": 95` on an untested equation are misleading. Derive ratings from counted peer reviews and display uncertainty (e.g. "3 reviews, unverified").

Evidence base for these cautions [Empirical, moderate confidence]. Variable-ratio rewards produce persistent responding (Ferster & Skinner 1957). Machine design can deliberately exploit this (Schüll 2012). Manipulative interface "dark patterns" are documented at scale (Mathur et al. 2019). The relationship between screen use and wellbeing is contested and often small in population studies. Design should therefore avoid known compulsion mechanics without claiming the app will "cure" anything.

PART III: The Bridge (structural analogy, not identity)

5. Where the two share mathematical form

Template element	Quantum instantiation (Part I)	Zero-Point instantiation (Part II)	Strength of analogy
Decay	Relaxation: P_1(t) ∝ e^{−t/T_1}, rate Γ_1	Demurrage: V = V₀e^{−λt}, t½ = ln2/λ	**Exact form** (first-order linear decay)
Unwanted occupation	Thermal occupation n̄; excited population P_1	Idle fraction φ_idle (share of balance idle > t_idle)	Form only
Damping / friction	Γ_φ, filtered noise J_eff(ω)	Friction dampeners, f, f_min	Loose: both reduce response, but by different mechanisms
Threshold monitor	Readout discrimination; leakage and error-burst detection	D_danger vs D_crit	Shared template (detect → intervene)

Template element	Quantum instantiation (Part I)	Zero-Point instantiation (Part II)	Strength of analogy
Bounded control	Bounded slew $\|\theta\| \le \theta_{max}$; shaped pulses	Gentle nudges; capped intervention rate	Shared template
Rest state	Ground state $\|0\rangle$ with $\frac{1}{2}\hbar\omega$ intact; cooling sends $\bar{n} \to 0$	Sanctuary: demurrage reduced ($\lambda \to \lambda_s$), activity paused	Meaningful parallel: **rest is not zero** in either case
No perfect isolation	Residual Γ always > 0	$\lambda_s > 0$, $f_{min} > 0$, so nothing diverges	Same design lesson
Optimum, not maximum	Faster gates increase leakage; t_g has a sweet spot	Velocity G_W has a healthy band; more churn ≠ better	Same design lesson

6. Where the analogy breaks (disanalogies to state openly)

1. **Agency and strategy.** Qubits do not respond to the metric. People do. Goodhart's law (Goodhart 1975) and the Lucas critique (Lucas 1976) warn that a measure used as a target changes behavior, so D_{danger}, ΔH, and patience weight will be gamed in ways T_1 never is.
2. **Consent and dignity.** Controlling a qubit raises no ethical question. Nudging a person does: interventions must be transparent, optional, and reversible.
3. **Units and physical law.** $\bar{n}$ follows from Bose–Einstein statistics. φ_{idle} follows from design choices and human behavior. There is no conservation law or thermodynamic temperature for Chrons unless one is defined.
4. **The zero-point resonance is poetic.** "Rest is not nothing" is true of a quantum ground state and is a good values statement about Sanctuary. The physics does not prove the values claim.

7. The merged master control objective (CUD template)

```
GENERAL FORM
  max over u(·)   J[u] = Q(u) − τ · Σ_k Γ_k(u, env) − ε_ctrl[u]
  s.t.            n_unwanted(u) ≤ n_max ;   R(u) ≤ R_max ;
                  rest state reachable at all times with residual decay Γ_rest > 0
(never exactly 0) ;
                  |u̇| ≤ u̇_max  (bounded, gentle control)
```

Symbol	Quantum instantiation	Zero-Point instantiation
$u(\cdot)$	{θ(t), T, □}: bias trajectory, temperature, shielding/filter stack	{λ, λ_s, f, nudge policy, D_{crit}}: app parameters
$Q(u)$	1 (ideal gate)	Band score: $Q = 1 - \|G_W - G\|/G$, with G^* a healthy target velocity from the pilot
τ	$t_g/3$	Window weight W (normalizes rates per window)
$\Sigma\ \Gamma_k$	$\Gamma_1 + \Gamma_\varphi$ (from $\bar{n}$, J_{eff}, Γ_P, Γ_{qp})	$w_1 \cdot \lambda \cdot \varphi_{idle} + w_2 \cdot \tilde{r}_{imp}$ (idle decay + impulsive rate)

Symbol	Quantum instantiation	Zero-Point instantiation
ε_ctrl	Leakage + non-adiabatic (LZ) error of θ(t)	Intervention burden: nudges per session, user-rated intrusiveness
n_unwanted ≤ n_max	$\bar{n}(\omega_q,T) \le n_{max}$	φ_idle ≤ φ_max
R ≤ R_max	Γ_env ≤ Γ_max	D_danger ≤ D_crit, **and** wellbeing (SWEMWBS) not declining vs baseline
Rest state	|0⟩, ½ħω intact	Sanctuary / off-app time, λ_s > 0
J measured as	Average gate fidelity $\bar{F}$ (randomized benchmarking)	Composite pilot score, reported as separate components (not just the sum)

Plain-text summary: `J = Q − τ·ΣΓ − ε_ctrl`. Quantum: $\bar{F} \approx 1 - (t_g/3)(\Gamma_1+\Gamma_\varphi) - \varepsilon_{ctrl}$. Zero-Point: `J_econ = [1 − |G_W − G*|/G*] − W·(w₁λφ_idle + w₂r̃_imp) − ε_nudge`, subject to `D_danger ≤ D_crit` and non-declining wellbeing.

PART IV: Testing, Limits, and Meaning

8. Falsifiability: quantum (brief)

1. **Coherence vs. temperature.** Residual P_1 and 1/T_1 should track $\bar{n}(\omega_q,T)$. Zero-point signatures (sideband asymmetry) must **persist** as $\bar{n} \to 0$.
2. **Coherence vs. filtering and shielding.** Improvements should scale with the reduction in J_eff(ω_q) and in radiation flux. If there is no change, Layer C is not the bottleneck.
3. **Landau–Zener sweeps.** Linear sweeps should follow P_LZ. Shaped θ(t) should beat linear sweeps at equal duration, or "smart gating" adds nothing.

9. Falsifiability: small user pilot for the app

Design [proposal]. A small, pre-registered feasibility pilot (e.g. a few dozen adult volunteers over about 4 weeks). It compares the full safety stack (Danger monitor, Sanctuary with off-app credit, no lottery) with the same app minus nudges, within-subject or as two arms. Use informed consent, data private by default, and ethics/IRB review if results will be published.

Metrics (all defined in Section 3):

- Circulation velocity G_W, and whether it settles into a band rather than spiking.
- Idle decay: φ_idle and Chrons recycled to the treasury.
- Impulsive purchase rate r_imp (operational definition fixed before launch).
- Sanctuary time and off-app time between sessions; session count and length.
- D_danger exceedances and the user's response (accept or dismiss reflection).
- Wellbeing: **SWEMWBS**, the 7-item short Warwick–Edinburgh Mental Well-being Scale (Tennant et al. 2007; Stewart-Brown et al. 2009). Measure at baseline and at

the end. Check its license terms; registration is required, and non-commercial use is generally free.
- ΔH audit: the fraction of ΔH-passing modules that human reviewers judge substantive. This measures how gameable the semantic gate is.

What would count against the design:

- The nudge arm shows more sessions or compulsive checking.
- Wellbeing declines relative to baseline or control.
- Impulsive rate does not fall.
- ΔH passes mostly padded edits.
- Demurrage drives users away instead of toward circulation.

What would support it: lower r_imp, stable G_W within a band, more off-app rest, and non-declining or improved SWEMWBS. All of this is reported as feasibility evidence, not proof.

10. What this is not

- **Not a theory of everything**, and not a claim that economics, psychology, or theology are quantum mechanics. Part III is a structural analogy.
- **Not zero-point energy harvesting or suppression.** ½ħω is irreducible. The "Zero-Point" name is a metaphor for rest that is not empty.
- **Not a modification of the cosmological constant.**
- **Not thermodynamics of money.** "Entropy" here means Shannon entropy of transactions, or it is metaphor.
- **Not a financial product, investment, or gambling offering.** Chrons are app credits. The lottery needs legal review or replacement.
- **Not a clinical or mental-health intervention.** The wellbeing metric is a research outcome, not a treatment claim.
- **Not a guarantee of equality.** Asymptotic equality is an aspiration to design toward and to measure.

11. Limitations

- The quantum fidelity model is first-order and Markovian. The economic objective has free weights (w_1, w_2, G*, D_crit) that must be calibrated and could be tuned to tell any story. Pre-registration guards against that.
- Small pilots detect large effects only, and self-selected users bias the results.
- Historical demurrage episodes (Wörgl) were brief and confounded by the Depression and by political intervention.
- The CUD template's value is organizational. It should be judged by whether it produces better qubit experiments and a safer, fairer app.

12. The values layer (kept, and kept distinct)

The **Circuit of Grace**, living with Source, Kairos rest, and Buber's contrast between I–Thou and I–It relationships (Buber 1923) are the reason this framework exists. They tell the optimizer **what counts as success**: non-extraction, dignity, honest rest, and freedom to leave. Science can test whether a design achieves those aims. It cannot certify or refute the aims themselves. Keeping the two layers distinct protects the faith from being judged as a failed experiment, and protects the experiments from being judged as preaching.

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All credit and glory to God. I owe my knowledge to my Lord Jesus Christ. AI assistant: Grok Bot.